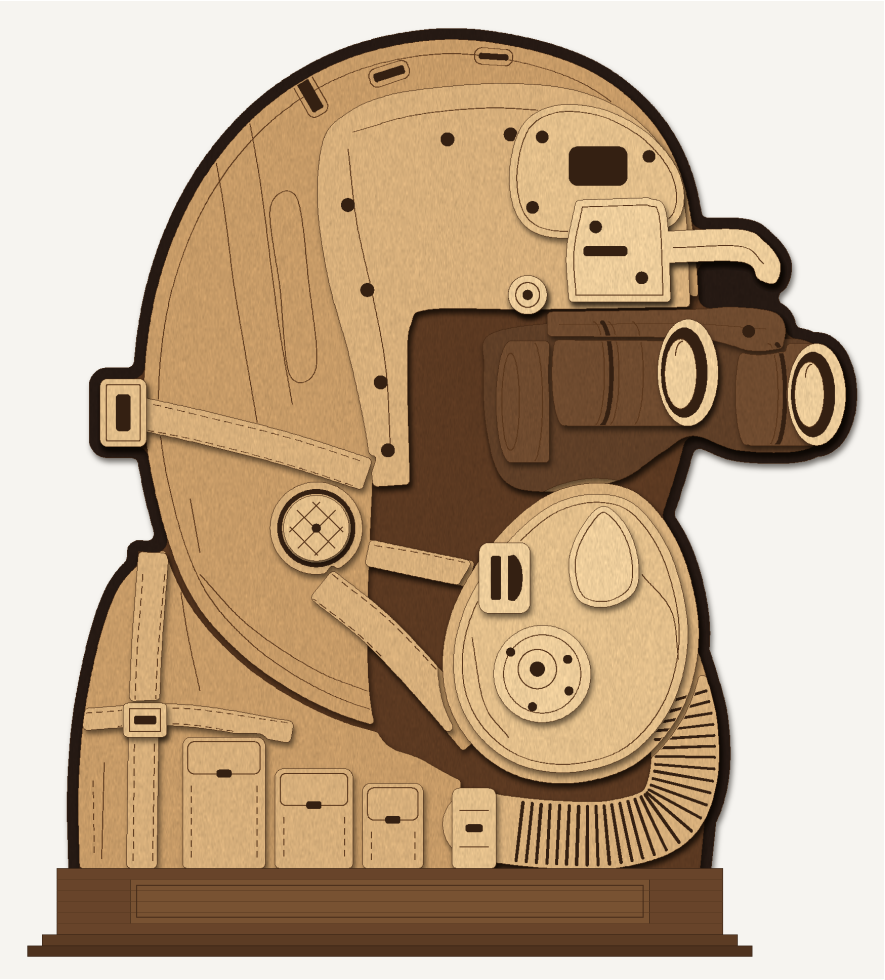
Pilot Helmet Layered Plaque

xTool S1 cut files | 6 plaque layers + 8-layer stand | 3 mm basswood / birch ply



Finished size

Plaque 215 x 237 mm, 18 mm thick
On stand 215 W x 261 H mm (about 8.4 x 10.3 in)
Stand footprint 197 x 64 mm

Material

3 mm basswood or birch ply.
Plaque: 6 sheets, 12 x 12 in (each layer file is about 225 x 247 mm).
Stand: 2 sheets, 12 x 12 in.
Wood glue, stain / paint, 220 grit sandpaper, tweezers.

Colour code in every SVG

RED = Cut
BLUE = Score
BLACK fill = Engrave
GREEN = placement guide (optional)
Green guides sit 0.8 mm inside where the next layer goes, so they end up hidden. Score them lightly, or delete them if you prefer to align by eye.

xTool Creative Space (XCS) workflow

1. Import one SVG at a time. Files are in millimetres at true size - do not rescale the backer or stand, or the tab and slot will not match. To make it bigger or smaller, scale all six plaque layers by the same percentage and resize the stand slot to the new tab width.
2. Ungroup, then set the process by colour: red = Cut, blue = Score, black = Engrave (fill), green = Score at low power (or delete). Start from the XCS material preset for 3 mm basswood and run a small test cut first.
3. All plaque files share the same canvas and origin, so every piece is already in its assembled position.
4. The stand slot is 3.3 mm wide for 3.0 mm stock. Measure your sheet; sand the tab if the fit is tight.

Layer stack (back to front)

Glue each layer onto the one before it. Finish suggestions match the reference look.



L1 backer

Backer / outline

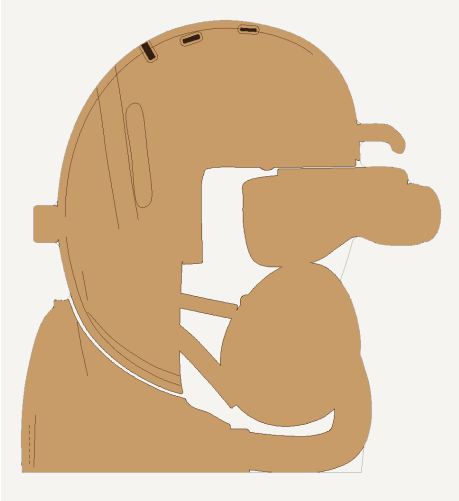
Finish: Black stain or paint (the outline)



L2 shadow

Shadow layer

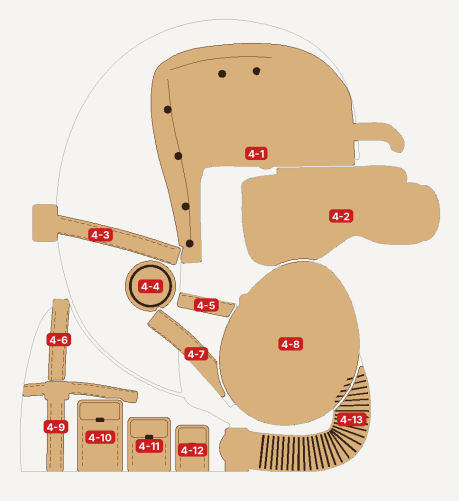
Finish: Dark walnut (shows in the visor opening and every gap)



L3 shell-and-vest

Helmet shell + flight vest

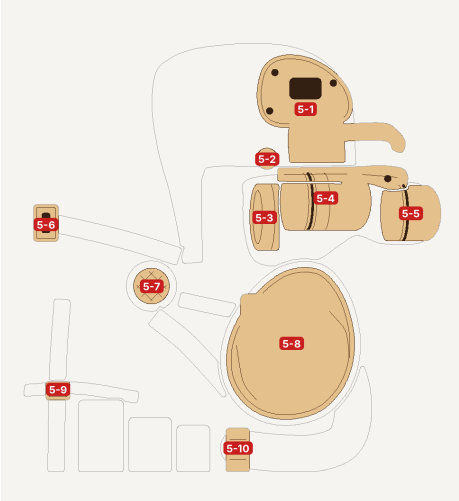
Finish: Natural or light honey



L4 housing-straps-mask

Visor housing, straps, ear cup, mask seal, NVG housing, hose, pouches

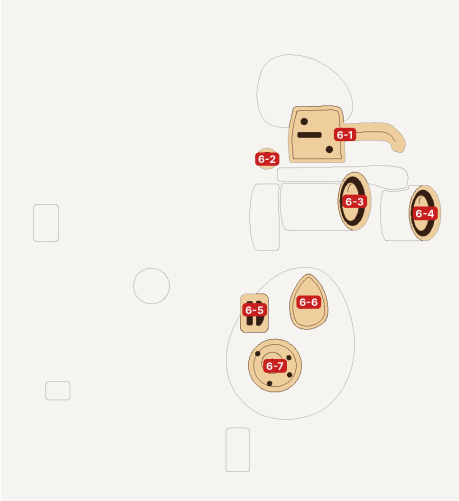
Finish: Natural; NVG housing in dark walnut



L5 nvg-tubes-mask-shell

NVG tubes, mount shroud & bracket, mask shell, buckles

Finish: Natural; NVG tubes + eyecup in dark walnut/espresso



L6 lenses-mount-valve

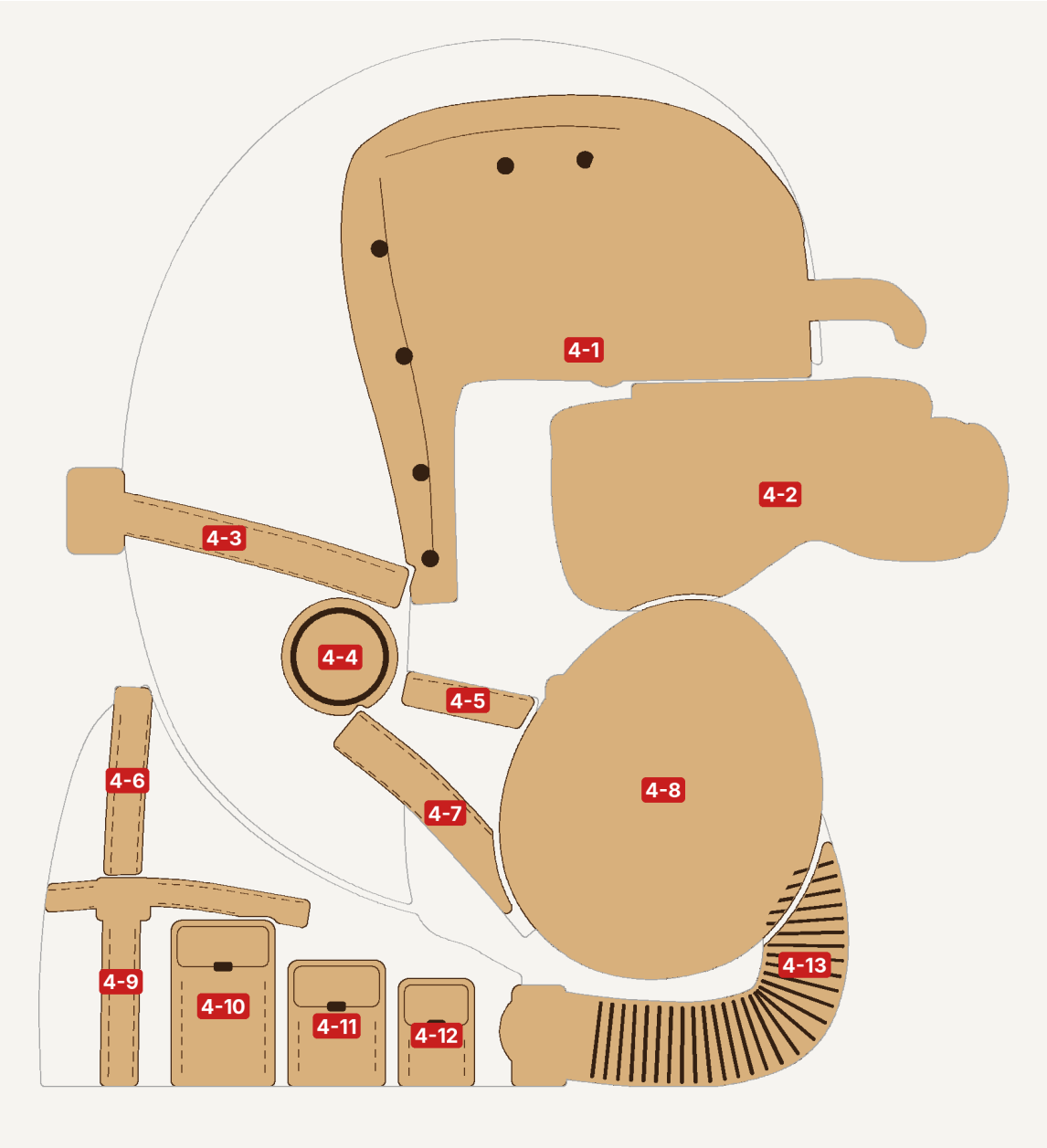
Objective lenses, NVG mount + tilt arm, knob, nose plate, valve, mask buckle

Finish: Natural (lightest pieces on top)

Grey outline = edge of the layer underneath. Layers 4-6 have numbered piece maps on the next pages.

Layer 4 pieces (13 pieces)

Visor housing, straps, ear cup, mask seal, NVG housing, hose, pouches



4-1 visor housing [126 x 109 mm]

4-2 NVG housing [99 x 50 mm] - dark stain

4-3 side strap [74 x 30 mm]

4-4 ear cup [25 x 25 mm]

4-5 mask strap [28 x 12 mm]

4-6 harness [11 x 41 mm]

4-7 chin strap [38 x 43 mm]

4-8 mask seal [70 x 82 mm]

4-9 harness, chest strap [57 x 45 mm]

4-10 pouch 1 [22 x 36 mm]

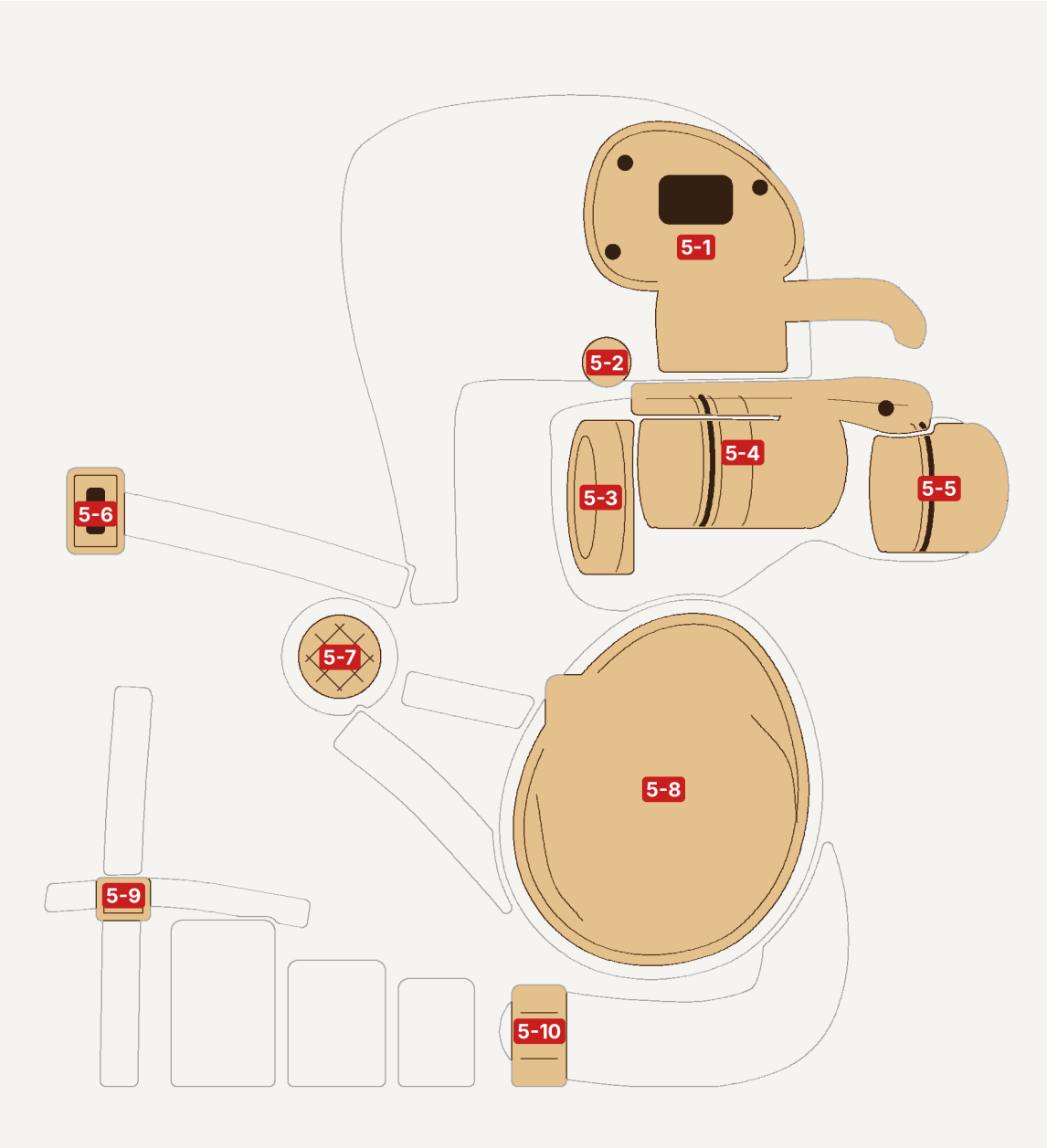
4-11 pouch 2 [21 x 27 mm]

4-12 pouch 3 [16 x 23 mm]

4-13 O2 hose [75 x 52 mm]

Layer 5 pieces (10 pieces)

NVG tubes, mount shroud & bracket, mask shell, buckles



5-1 mount shroud [74 x 54 mm]

5-2 base under mount knob [11 x 11 mm]

5-3 eyecup [15 x 33 mm] - dark stain

5-4 NVG tube L, mount bracket [65 x 32 mm] - dark stain

5-5 NVG tube R [30 x 28 mm] - dark stain

5-6 strap buckle [13 x 18 mm]

5-7 ear cap [18 x 18 mm]

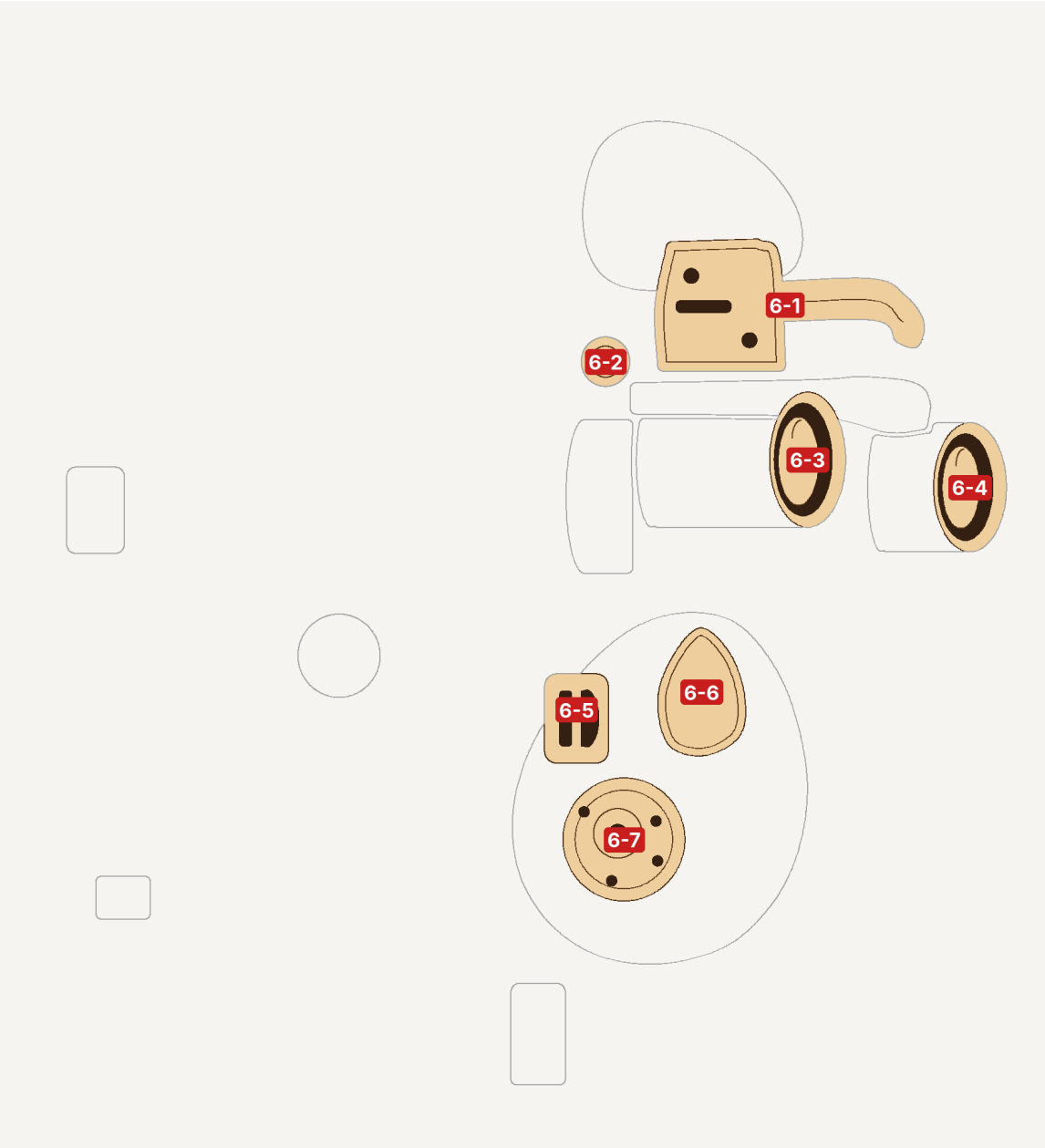
5-8 mask shell [64 x 76 mm]

5-9 harness buckle [12 x 9 mm]

5-10 hose fitting [12 x 22 mm]

Layer 6 pieces (7 pieces)

Objective lenses, NVG mount + tilt arm, knob, nose plate, valve, mask buckle



6-1 mount + tilt arm [58 x 28 mm]

6-2 mount knob [11 x 11 mm]

6-3 lens L [16 x 29 mm]

6-4 lens R [16 x 28 mm]

6-5 mask buckle [14 x 19 mm]

6-6 nose plate [19 x 27 mm]

6-7 mic / valve [26 x 26 mm]

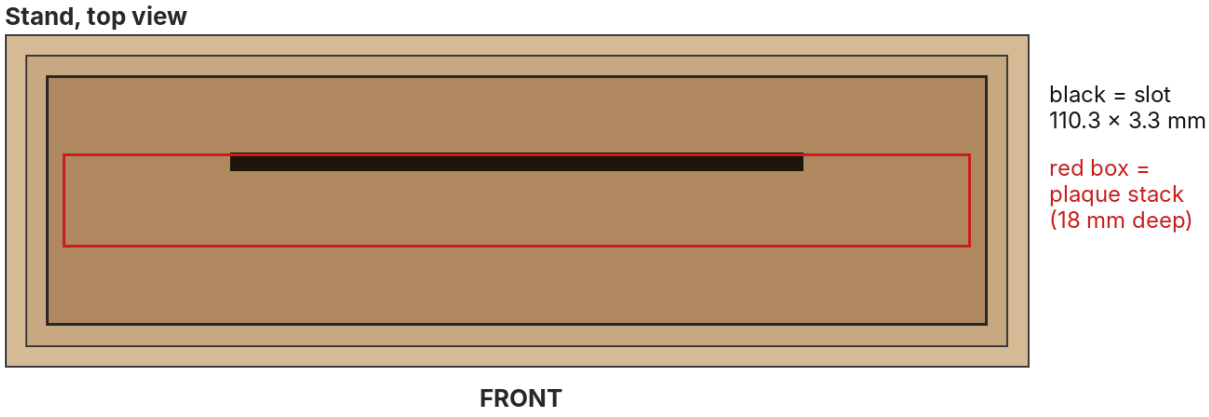
Assembly & stand

Plaque

1. Cut every layer. Lightly sand the top faces to remove smoke residue but keep the charred edges - they make the dark outlines. Stain or paint each layer before gluing (page 2).
2. Glue L2 onto L1 so the black border shows evenly all round. The bottom edges line up flat.
3. Glue L3 inside the green guide on L2, then keep going upward - every layer carries a green guide for the next one. For small pieces use a toothpick of glue in the centre and place with tweezers.
4. Press the finished stack under a flat weight for 1-2 hours.

Stand

5. Glue B2 centred on B1 (green guide), then B3 to B8 squarely on top using the guide on B2. The slot in B6-B8 sits 30 mm from the FRONT edge (closer to the back) - keep the three slots lined up.
6. Optional: add a name or callsign to the nameplate in XCS before cutting, then glue it centred on the front of the block.
7. Push the plaque tab into the slot with a little glue. The bottom of the whole stack rests on the stand.



Stand layers (bottom to top)

B1 plinth 197 x 64 mm | B2 plinth step 189 x 56 mm | B3-B5 block 181 x 48 mm | B6-B8 block with 110.3 x 3.3 mm slot
Plaque tab 110 x 8.6 mm drops into the 9 mm deep slot. Nameplate 141 x 12 mm fits the 18 mm tall block front. File B-A = B1-B4, file B-B = B5-B8 + nameplate (each fits a 12 x 12 in sheet). Dark walnut on the whole stand matches the reference.